

Algebra and Discrete Mathematics

ADM_A_B

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Course Outline

- Vectors and matrices
- System of linear equations
- Matrix inverse and determinants
- Vector spaces, bases and matrix transformations
- Quiz one
- Matrix transformations, fundamental spaces and decompositions
- Eulerian tours and Hamiltonian cycles
- Paths and spanning trees
- Quiz two
- Trees and networks
- Matching

Recommended reading

- Andrilli, Stephen, and David Hecker. Elementary linear algebra. Academic Press, 2022. Fifth edition
 - Sections 1.1, 1.2, 1.4, 1.5
 - [Accessible online \(free copy\)](#)
 - [Alternative download link](#)
- Anton, Howard, and Chris Rorres. Elementary linear algebra: applications version. John Wiley & Sons, 2013.
 - Sections 1.3
 - [Accessible online \(free copy\)](#)
 - [Alternative download link](#)

Lecture outline

- Definitions
- Vectors
- Matrices
- Proofs and principles

Vectors and matrices

- Definitions
- Vectors
- Matrices
- Proofs and principles

Sets

- *set*: an unordered collection of distinct objects
- $\emptyset$: empty set
- $|S|$: cardinality of S
- A set S is *finite* if $|S| < \infty$
- $a \in S$: a is an element of S
- $a \notin S$: a is not an element in set S
- $S \subseteq T$: if $s \in S$, then $s \in T$, S is a subset of T
- $S = T$: $S \subseteq T$ and $T \subseteq S$
- The *power set* of a set S , denoted by 2^S , is the set of all subsets of S .

Example

Let $T = \{0, 1, 2, 3\}$ and $S = \{2, 3\}$, then

- $S \subseteq T$ and $T \not\subseteq S$.
- $2 \in S$, $0 \notin S$.
- $|S| = 2$, $|T| = 4$.
- $2^S = \{\emptyset, S, \{2\}, \{3\}\}$.

Sets

- Union: $A \cup B$
- Intersection: $A \cap B$
- Difference: $A \setminus B = \{ a \in A \mid a \notin B \}$
- Complement of A in S : $A^c = S \setminus A$
- Cartesian product $A \times B = \{ (a, b) \mid a \in A, b \in B \}$
 - ordered pairs

Example

- $A = \{ 0, 1, 2 \}, B = \{ 2, 3, 4 \}$
- $A \cup B = \{ 0, 1, 2, 3, 4 \}, A \cap B = \{ 2 \}$

Example

- $A = \{ 2, 4, 6 \}, B = \{ 1, 3, 5 \}, S = A \cup B$
- $A \setminus B = A$. Complement of A in S is B

$$A \times B = \{ (2, 1), (2, 3), (2, 5), (4, 1), (4, 3), (4, 5), (6, 1), (6, 3), (6, 5) \}.$$

Functions

Definition

Suggested: A *function* (or *map*) $f : S \rightarrow T$ is a rule that assigns to each $s \in S$ exactly one element $t \in T$.

- S – *domain* of f ; T – *codomain* of f .
- If $f(s) = t$, then t is called the *image* of s , s is a *preimage* of t .
- For any $A \subseteq T$, *preimage of A under f* is

$$f^{-1}(A) := \{ s \in S \mid f(s) \in A \}$$

Example

Define

$$\begin{aligned} f : \mathbb{R} &\rightarrow \mathbb{R} \\ x &\mapsto x^2 \end{aligned}$$

where $\mathbb{R}$ is the set of real numbers. Then f has domain $\mathbb{R}$ and codomain $\mathbb{R}$.

Functions – example

Example

Define

$$\begin{aligned} f : \mathbb{R} &\rightarrow \mathbb{R} \\ x &\mapsto x^2 \end{aligned}$$

- f has domain $\mathbb{R}$ and codomain $\mathbb{R}$
- Let $A = \{ 1 \} \subseteq \mathbb{R}$, the preimage of A under f is given by

$$f^{-1}(A) = \{ -1, 1 \}.$$

- 1 is the image of -1 and -1 is a preimage of 1.
- 1 is another preimage of 1.
- Let $B = \{ -1 \} \subseteq \mathbb{R}$, then $f^{-1}(B) = \emptyset$.

Functions

Definition

- A function $f : S \rightarrow T$ is called *onto* or *surjective* if given any $t \in T$, there exists $s \in S$, such that $t = f(s)$.
- A function $f : S \rightarrow T$ is said to be *one-to-one* (written 1-1) or *injective* if for any $s_1, s_2 \in S$ such that $s_1 \neq s_2$, we have $f(s_1) \neq f(s_2)$.
- f is called *1-1 correspondence* or *bijective* if f is 1-1 and onto.

Example

Is f injective, surjective, or bijective? Is g injective, surjective, or bijective?

$$\begin{aligned} f : \mathbb{R} &\rightarrow \mathbb{R}_{\geq 0} \\ x &\mapsto x^2 \end{aligned}$$

$$\begin{aligned} g : \mathbb{R} &\rightarrow \mathbb{R} \\ x &\mapsto x \end{aligned}$$

Functions

Example

$$\begin{aligned} f : \mathbb{R} &\rightarrow \mathbb{R}_{\geq 0} \\ x &\mapsto x^2, \end{aligned}$$

- f is surjective as for any $y \in \mathbb{R}_{\geq 0}$, we can find a preimage of y by calculating $x = \sqrt{y}$.
- But f is not injective, since $f(-1) = f(1) = 1$.

$$\begin{aligned} g : \mathbb{R} &\rightarrow \mathbb{R} \\ x &\mapsto x. \end{aligned}$$

- It can be easily seen that g is bijective

Inverse of a function

- When $f : S \rightarrow T$ is bijective, $f^{-1} : T \rightarrow S$ is a function – it assigns each $t \in T$ a unique element $s \in S$.
- f^{-1} is called the *inverse* of f .

Example

Define f

$$\begin{aligned} f : \mathbb{R} &\rightarrow \mathbb{R} \\ x &\mapsto x^3. \end{aligned}$$

Then, the inverse of f exists and is given by

$$\begin{aligned} f^{-1} : \mathbb{R} &\rightarrow \mathbb{R} \\ x &\mapsto \sqrt[3]{x}. \end{aligned}$$

Composition of functions

Definition

For two functions $f : T \rightarrow U$, $g : S \rightarrow T$, the *composition* of f and g , denoted by $f \circ g$, is the function

$$\begin{aligned} f \circ g : S &\rightarrow U \\ s &\mapsto f(g(s)). \end{aligned}$$

Example

What is $f \circ g$?

$$\begin{aligned} f : \mathbb{R} &\rightarrow \mathbb{R} \\ x &\mapsto x^2, \end{aligned}$$

$$\begin{aligned} g : \mathbb{R} &\rightarrow \mathbb{R} \\ x &\mapsto x^3. \end{aligned}$$

Composition of functions

Example

$$\begin{aligned} f : \mathbb{R} &\rightarrow \mathbb{R} \\ x &\mapsto x^2, \end{aligned}$$

$$\begin{aligned} g : \mathbb{R} &\rightarrow \mathbb{R} \\ x &\mapsto x^3. \end{aligned}$$

$$\begin{aligned} f \circ g : \mathbb{R} &\rightarrow \mathbb{R} \\ x &\mapsto (x^3)^2 = x^6. \end{aligned}$$

Matrices

- $\mathbb{R}$: the set of all real numbers

Definition

A *matrix with coefficients in $\mathbb{R}$* is a rectangular array where each entry is an element of $\mathbb{R}$.

Matrix A is said to have m rows, n columns and is of size $m \times n$.

$$A = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ a_{21} & \dots & a_{2n} \\ & \vdots & \\ a_{m1} & \dots & a_{mn} \end{pmatrix}.$$

Example

The matrix

$$A = \begin{pmatrix} 1 & 7.5 & 6 \\ 2 & 3 & 4 \end{pmatrix}$$

has 2 rows, 3 columns and is of size 2×3 .

Vectors

- A $1 \times n$ matrix is called a *row vector*.
- An $n \times 1$ matrix is called a *column vector*.

Example

- $\mathbf{a} = (1, -1, 3)$ is a row vector
- $\mathbf{b} = \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}$ is a column vector

Note

- By “vector,” we normally refer specifically to a row vector.
- $\mathbb{R}^n$: the set of all vectors with n entries, also referred to as *coordinates*.
- When written by hand, $\vec{a}$ is used to denote a vector.

Example

$$\mathbf{a} \in \mathbb{R}^3$$

Vectors and scalars

- Two vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ are equal, written $\mathbf{a} = \mathbf{b}$, if all corresponding coordinates are *equal*
- $\mathbf{0} = (0, 0, \dots, 0)$ is the *zero vector*.
- An element $x \in \mathbb{R}$ is called a *scalar*

Example

- $(1, 0, 4) \neq (1, 0, -4)$
- $5 \in \mathbb{R}$ is a scalar

If and only if

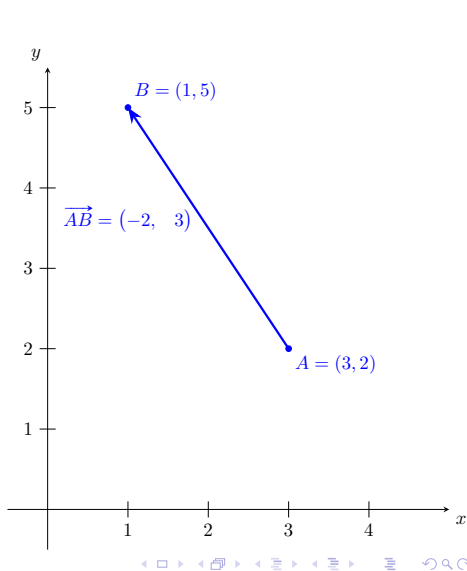
- In short: p iff q , or $p \Leftrightarrow q$
- The condition laid out are both necessary and sufficient, q is a necessary and sufficient condition for p
- *Necessary condition*: $p \Rightarrow q$, q is a necessary condition for p , a property that must be achieved in order for p to be true
- *Sufficient condition*: $p \Leftarrow q$, q is a sufficient condition for p , property that guarantees that p is true

Vectors and matrices

- Definitions
- **Vectors**
- Matrices
- Proofs and principles

Geometric interpretation of vectors

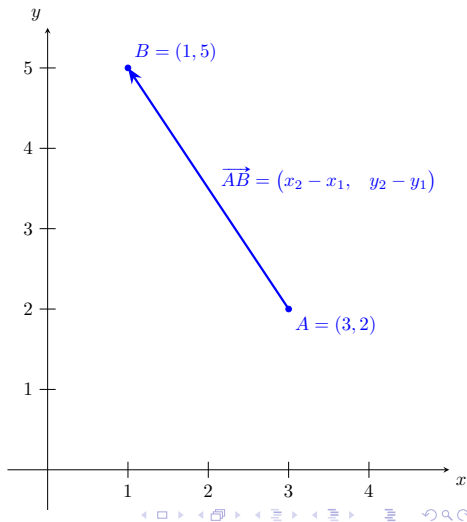
- A vector with two coordinates, i.e. an element of $\mathbb{R}^2$, is frequently used to represent a movement from one point to another in a coordinate plane
- From an initial point $(3, 2)$ to a terminal point $(1, 5)$, there is a net decrease of 2 units along the x -axis and a net increase of 3 units along the y -axis. A vector representing this change would thus be $(-2, 3)$, as indicated by the arrow in the figure



Geometric interpretation of vectors

- In general, a vector starting at point $A = (x_1, y_1)$ and ending at $B = (x_2, y_2)$, denoted $\overrightarrow{AB}$ is given by

$$\overrightarrow{AB} = (x_2 - x_1, y_2 - y_1)$$



Norm of a vector

- The *distance* between two points (x_1, y_1) and (x_2, y_2) in the plane is given by

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

- The vector between the points is $(x_2 - x_1, y_2 - y_1)$
- This motivates the following definition

Definition

The *norm* (also called *length*) of a vector $\mathbf{a} = (a_1, a_2, \dots, a_n)$, denoted $\|\mathbf{a}\|$, is given by

$$\|\mathbf{a}\| = \sqrt{a_1^2 + a_2^2 + \dots + a_n^2}.$$

A vector of norm 1 is called a *unit vector*

Norm of a vector – Example

Example

- The norm of $\mathbf{a} = (4, -3, 0, 2)$ is given by

$$\|\mathbf{a}\| = \sqrt{4^2 + (-3)^2 + 0^2 + 2^2} = \sqrt{16 + 9 + 0 + 4} = \sqrt{29}$$

- $\left(\frac{3}{5}, -\frac{4}{5}\right)$ is a unit vector in $\mathbb{R}^2$

$$\sqrt{\left(\frac{3}{5}\right)^2 + \left(-\frac{4}{5}\right)^2} = 1$$

- $\left(-\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2}\right)$ is a unit vector in $\mathbb{R}^4$

$$\sqrt{\left(-\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 + \left(-\frac{1}{2}\right)^2} = 1$$

Scalar multiplication

- $\mathbf{a} = (a_1, a_2, \dots, a_n) \in \mathbb{R}^n$
- $\alpha \in \mathbb{R}$
- The scalar multiple of $\mathbf{a}$ by α is the vector

$$\alpha \mathbf{a} = (\alpha a_1, \alpha a_2, \dots, \alpha a_n)$$

- It is easy to see that

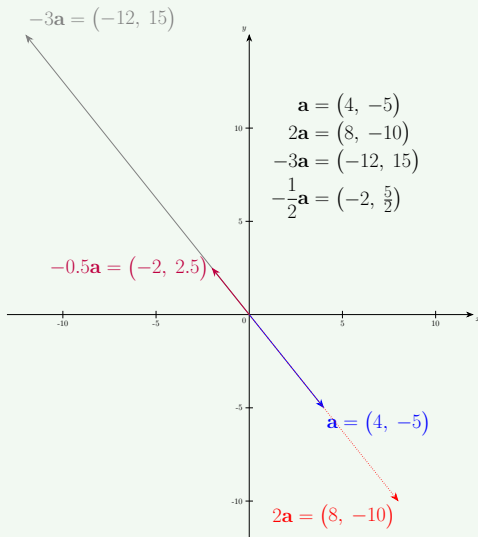
$$\|\alpha \mathbf{a}\| = |\alpha| \|\mathbf{a}\|$$

since

$$\|\alpha \mathbf{a}\| = \sqrt{(\alpha a_1)^2 + (\alpha a_2)^2 + \dots + (\alpha a_n)^2} = \sqrt{\alpha^2(a_1^2 + a_2^2 + \dots + a_n^2)} = |\alpha| \|\mathbf{a}\|.$$

Scalar multiplication

Example



Scalar multiplication

$$\|\alpha \mathbf{a}\| = |\alpha| \|\mathbf{a}\|$$

- Multiplication by α dilates (expands) the norm of the vector when $|\alpha| > 1$ and contracts (shrinks) the norm when $|\alpha| < 1$
- Scalar multiplication by 1 or -1 does not affect the norm
- Scalar multiplication by 0 always yields the zero vector.

Direction

Definition

$\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$, $\mathbf{a} \neq \mathbf{0}$, $\mathbf{b} \neq \mathbf{0}$, $\mathbf{a}$ and $\mathbf{b}$ are said to be

- *in the same direction* if $\exists \alpha \in \mathbb{R}_{>0}$ s.t. $\mathbf{b} = \alpha \mathbf{a}$
- *in the opposite direction* if $\exists \alpha \in \mathbb{R}_{<0}$ s.t. $\mathbf{b} = \alpha \mathbf{a}$
- *parallel* if they are in the same or in the opposite direction

Example

- $(1, -3, 2)$ and $(3, -9, 6)$ are in the same direction

$$(1, -3, 2) = \frac{1}{3}(3, -9, 6).$$

- $(-3, 6, -15)$ and $(4, -8, 20)$ are in the opposite direction

$$(-3, 6, -15) = -\frac{3}{4}(4, -8, 20).$$

Normalization of a vector

Lemma

For any $\mathbf{a} \in \mathbb{R}^n$, $\mathbf{a} \neq \mathbf{0}$,

$$\frac{\mathbf{a}}{\|\mathbf{a}\|}$$

is a unit vector in the same direction as $\mathbf{a}$.

Proof.

By the above observations

$$\left\| \frac{\mathbf{a}}{\|\mathbf{a}\|} \right\| = \frac{1}{\|\mathbf{a}\|} \|\mathbf{a}\| = 1.$$



This process of “dividing” a vector by its norm to obtain a unit vector in the same direction is called *normalizing* the vector.

Vector addition

- Take two vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$

$$\mathbf{a} = (a_1, a_2, \dots, a_n), \quad \mathbf{b} = (b_1, b_2, \dots, b_n)$$

- The sum of $\mathbf{a}$ and $\mathbf{b}$ is given by the vector

$$(a_1 + b_1, a_2 + b_2, \dots, a_n + b_n)$$

Example

$$\mathbf{a} = (2, -3, 5), \mathbf{b} = (-6, 4, -2),$$

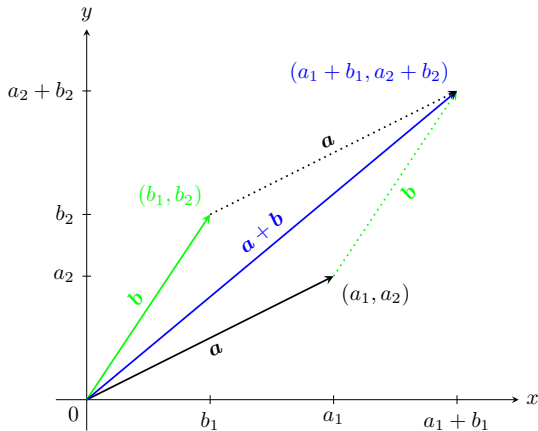
$$\mathbf{a} + \mathbf{b} = (2 - 6, -3 + 4, 5 - 2) = (-4, 1, 3)$$

Note

Vectors cannot be added unless they have the same number of coordinates

Vector addition – geometric interpretation

- Draw a vector $\mathbf{a}$. Then draw a vector $\mathbf{b}$ whose initial point is the terminal point of $\mathbf{a}$.
- The sum of $\mathbf{a}$ and $\mathbf{b}$ is the vector whose initial point is the same as that of $\mathbf{a}$ and whose terminal point is the same as that of $\mathbf{b}$.
- The total movement $\mathbf{a} + \mathbf{b}$ is equivalent to first moving along $\mathbf{a}$ and then along $\mathbf{b}$



Subtraction of vectors

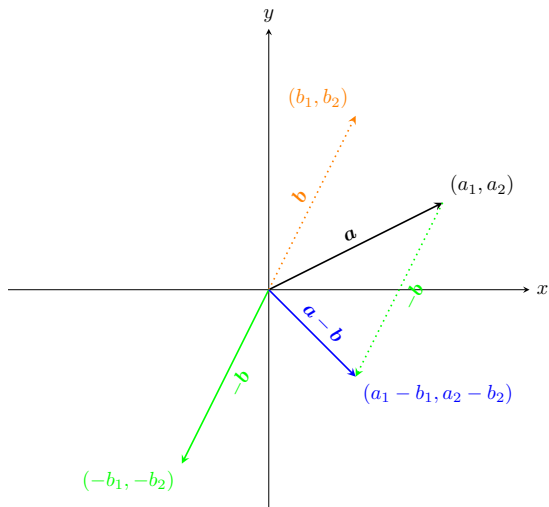
- Let $-b$ denote the scalar product between -1 and b
- Define

$$a - b = a + (-b)$$

Example

$$a = (2, 1), b = (1, 2)$$

$$\begin{aligned} a - b &= a + (-b) \\ &= (2, 1) + (-1, -2) \\ &= (1, -1). \end{aligned}$$



Fundamental properties of vector addition and scalar multiplication

Theorem

Take any $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^n$, any $\alpha, \beta \in \mathbb{R}$, we have

- | | |
|--|--|
| 1. $\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a}$ | <i>Commutative law of addition</i> |
| 2. $\mathbf{a} + (\mathbf{b} + \mathbf{c}) = (\mathbf{a} + \mathbf{b}) + \mathbf{c}$ | <i>Associative law of addition</i> |
| 3. $\mathbf{0} + \mathbf{a} = \mathbf{a} + \mathbf{0} = \mathbf{a}$ | <i>Existence of identity element for addition</i> |
| 4. $\mathbf{a} + (-\mathbf{a}) = (-\mathbf{a}) + \mathbf{a} = \mathbf{0}$ | <i>Existence of inverse elements for addition</i> |
| 5. $\alpha(\mathbf{a} + \mathbf{b}) = \alpha\mathbf{a} + \alpha\mathbf{b}$ | <i>Distributive laws of scalar multiplication</i> |
| 6. $(\alpha + \beta)\mathbf{a} = \alpha\mathbf{a} + \beta\mathbf{a}$ | <i>over vector addition</i> |
| 7. $(\alpha\beta)\mathbf{a} = \alpha(\beta\mathbf{a})$ | <i>Associativity of scalar multiplication</i> |
| 8. $1\mathbf{a} = \mathbf{a}$ | <i>Identity property for scalar multiplication</i> |

- $\mathbf{0}$ is called an *identity element for vector addition* because $\mathbf{0}$ does not change the identity of any vector to which it is added
- $-\mathbf{a}$ is called the *Additive inverse of $\mathbf{a}$* because it “cancels out $\mathbf{a}$ ” to produce the additive identity element (i.e. the zero vector)

Dot product

Definition

Let

$$\mathbf{a} = (a_1, a_2, \dots, a_n), \mathbf{b} = (b_1, b_2, \dots, b_n) \in \mathbb{R}^n$$

be two vectors. The *dot product* (*inner product*) of $\mathbf{a}$ and $\mathbf{b}$ is given by

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + \dots + a_n b_n = \sum_{i=1}^n a_i b_i.$$

Example

$$\mathbf{a} = (2, -4, 3), \mathbf{b} = (1, 5, -2), \mathbf{a} \cdot \mathbf{b} = 2 \times 1 + (-4) \times 5 + 3 \times (-2) = -24.$$

Note

- Dot product is not defined for vectors having different numbers of coordinates.
- Dot product involves two vectors and the result is a scalar, whereas scalar multiplication involves a scalar and a vector and the result is a vector.

Properties of dot product

Theorem

Take any $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^n$, $\alpha \in \mathbb{R}$, then

1. $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$ *Commutativity of dot product*
2. $\mathbf{a} \cdot \mathbf{a} = \|\mathbf{a}\|^2 \geq 0$ *Relationship between dot product and norm*
3. $\mathbf{a} \cdot \mathbf{a} = 0$ iff $\mathbf{a} = \mathbf{0}$
4. $\alpha(\mathbf{a} \cdot \mathbf{b}) = (\alpha\mathbf{a}) \cdot \mathbf{b} = \mathbf{a} \cdot (\alpha\mathbf{b})$ *Relationship between scalar multiplication and dot product*
5. $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$ *Distributive laws of dot product*
6. $(\mathbf{a} + \mathbf{b}) \cdot \mathbf{c} = (\mathbf{a} \cdot \mathbf{c}) + (\mathbf{b} \cdot \mathbf{c})$ *over addition*

Proof.

$$2. \mathbf{a} = (a_1, a_2, \dots, a_n)$$

$$\mathbf{a} \cdot \mathbf{a} = a_1a_1 + a_2a_2 + \dots + a_na_n = a_1^2 + a_2^2 + \dots + a_n^2 = \|\mathbf{a}\|^2 \geq 0$$

Dot product – example

Example

$$\mathbf{a} = (1, 2, 3), \quad \mathbf{b} = (4, 5, 6), \quad \mathbf{c} = (-1, -2, -3), \quad \alpha = 2$$

$$\begin{aligned}\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) &= (1, 2, 3) \cdot ((4 - 1, 5 - 2, 6 - 3)) = (1, 2, 3) \cdot (3, 3, 3) \\ &= 1 \times 3 + 2 \times 3 + 3 \times 3 = 3 + 6 + 9 = 18,\end{aligned}$$

$$\mathbf{a} \cdot \mathbf{b} = (1, 2, 3) \cdot (4, 5, 6) = 1 \times 4 + 2 \times 5 + 3 \times 6 = 4 + 10 + 18 = 32,$$

$$\begin{aligned}\mathbf{a} \cdot \mathbf{c} &= (1, 2, 3) \cdot (-1, -2, -3) = 1 \times (-1) + 2 \times (-2) + 3 \times (-3) \\ &= -1 - 4 - 9 = -14,\end{aligned}$$

$$\mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c} = 32 - 14 = 18.$$

$$\alpha(\mathbf{a} \cdot \mathbf{b}) = 2 \times 32 = 64$$

$$(\alpha \mathbf{a}) \cdot \mathbf{b} = (2, 4, 6) \cdot (4, 5, 6) = 8 + 20 + 36 = 64$$

Orthogonal vectors

Definition

$\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ are *orthogonal* if $\mathbf{a} \cdot \mathbf{b} = 0$

Example

$\mathbf{a} = (2, -5)$ and $\mathbf{b} = (-10, -4)$ are orthogonal in $\mathbb{R}^2$

$$\mathbf{a} \cdot \mathbf{b} = 2 \times (-10) + (-5) \times (-4) = -20 + 20 = 0.$$

Dot product of unit vectors

Recall

A vector of norm 1 is called a *unit vector*

Lemma

If $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ are unit vectors, then

$$-1 \leq \mathbf{a} \cdot \mathbf{b} \leq 1.$$

Cauchy-Schwarz Inequality

Theorem

Take any $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$, we have

$$|\mathbf{a} \cdot \mathbf{b}| \leq \|\mathbf{a}\| \|\mathbf{b}\|.$$

Proof.

- If $\mathbf{a} = \mathbf{0}$ or $\mathbf{b} = \mathbf{0}$, the theorem holds
- Otherwise, the theorem is equivalent to

$$-1 \leq \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} \leq 1$$

We have discussed that

$$\frac{\mathbf{a}}{\|\mathbf{a}\|} \quad \frac{\mathbf{b}}{\|\mathbf{b}\|}$$

are unit vectors. The result follows from the previous lemma. □

Cauchy-Schwarz Inequality – example

Example

$$\mathbf{a} = (-1, 4, 2, 0, -3), \mathbf{b} = (2, 1, -4, -1, 0).$$

$$\mathbf{a} \cdot \mathbf{b} = -2 + 4 - 8 + 0 + 0 = -6$$

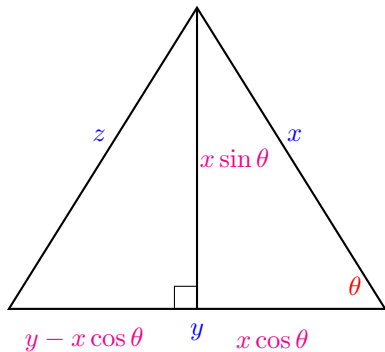
$$\|\mathbf{a}\| = \sqrt{1 + 16 + 4 + 0 + 9} = \sqrt{30}$$

$$\|\mathbf{b}\| = \sqrt{4 + 1 + 16 + 1 + 0} = \sqrt{22}$$

$$\|\mathbf{a}\| \|\mathbf{b}\| = \sqrt{30 \times 22} = 2\sqrt{165} \approx 25.7$$

$$|\mathbf{a} \cdot \mathbf{b}| = 6 \leq \|\mathbf{a}\| \|\mathbf{b}\|$$

Law of Cosines

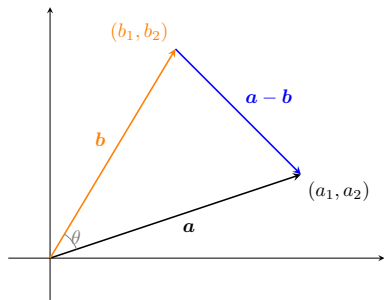


$$\begin{aligned}(y - x \cos \theta)^2 + (x \sin \theta)^2 &= z^2 \\ y^2 + x^2 \cos^2 \theta - 2yx \cos \theta + x^2 \sin^2 \theta &= z^2 \\ x^2 + y^2 - 2yx \cos \theta &= z^2\end{aligned}$$

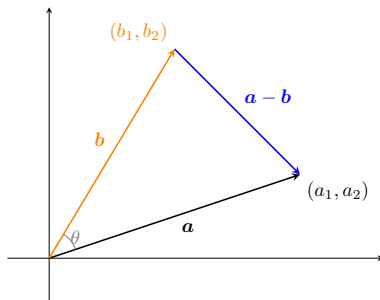
The angle between two vectors

- There are two angles formed by the two vectors, but we always choose the angle θ between two vectors to be the one measuring between 0 and π radians, inclusive.
- By the Law of Cosines

$$\|a-b\|^2 = \|a\|^2 + \|b\|^2 - 2\|a\|\|b\|\cos\theta$$



Angle between two vectors



$$\|a - b\|^2 = \|a\|^2 + \|b\|^2 - 2\|a\|\|b\|\cos\theta$$

$$\|a - b\|^2 = (a - b) \cdot (a - b) = a \cdot a - 2a \cdot b + b \cdot b = \|a\|^2 - 2a \cdot b + \|b\|^2$$

$$\implies \|a\|\|b\|\cos\theta = a \cdot b$$

$$\cos\theta = \frac{a \cdot b}{\|a\|\|b\|}$$

Angle between two vectors – example

Example

$$\mathbf{a} = (6, -4), \mathbf{b} = (-2, 3)$$

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} = \frac{6 \times (-2) + (-4) \times 3}{\sqrt{36 + 16} \sqrt{4 + 9}} = -\frac{24}{\sqrt{52} \sqrt{13}} = -\frac{12}{13} \approx -0.9231,$$

which gives $\theta \approx 2.74$ radians (using calculator).

Angle between two vectors

- For higher dimensions we are outside the geometry of everyday experience
- We give the following definition

Definition

For any $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ with $n \geq 2$, the *angle between $\mathbf{a}$ and $\mathbf{b}$* is the unique angle θ such that $0 \leq \theta \leq \pi$ and

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}.$$

Note that according to Cauchy-Schwarz Inequality,

$$-1 \leq \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} \leq 1.$$

Thus this value equals $\cos \theta$ for a unique θ from 0 to π radians.

Angle between two vectors

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}.$$

By the properties of the cosine function, we have

$$\mathbf{a} \cdot \mathbf{b} > 0 \iff 0 \leq \theta < \frac{\pi}{2}$$

$$\mathbf{a} \cdot \mathbf{b} = 0 \iff \theta = \frac{\pi}{2}$$

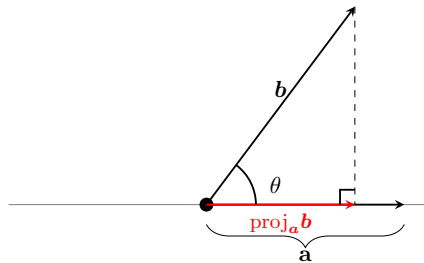
$$\mathbf{a} \cdot \mathbf{b} < 0 \iff \frac{\pi}{2} < \theta \leq \pi$$

Note

By definition of orthogonal vectors, two *nonzero* vectors are orthogonal if and only if they are perpendicular to each other (i.e. $\theta = \frac{\pi}{2}$)

Projection vectors

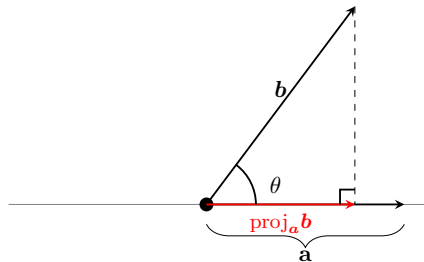
- The projection of one vector onto another is useful in physics, engineering, computer graphics, and statistics.
- Let $\mathbf{a}$, $\mathbf{b}$ be nonzero vectors, both in $\mathbb{R}^2$ or both in $\mathbb{R}^3$, drawn at the same initial point
- Let θ represent the angle between $\mathbf{a}$ and $\mathbf{b}$
- Drop a perpendicular line segment from the terminal point of $\mathbf{b}$ to the straight line containing the vector $\mathbf{a}$
- The projection of $\mathbf{b}$ onto $\mathbf{a}$, denoted $\text{proj}_{\mathbf{a}} \mathbf{b}$, is the vector from the initial point of $\mathbf{a}$ to the point where the dropped perpendicular meets the straight line



Projection vectors

- Using trigonometry, for $0 \leq \theta \leq \frac{\pi}{2}$, $\text{proj}_a \mathbf{b}$ is in the direction of the unit vector $\mathbf{a}/\|\mathbf{a}\|$, and

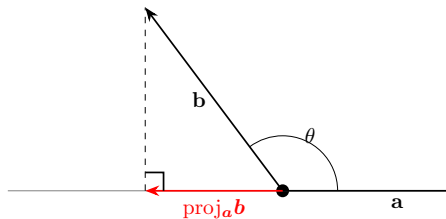
$$\|\text{proj}_a \mathbf{b}\| = \|\mathbf{b}\| \cos \theta$$



Projection vectors

- Using trigonometry, when $\frac{\pi}{2} < \theta \leq \pi$, $\text{proj}_a \mathbf{b}$ is in the direction of the unit vector $-\mathbf{a}/\|\mathbf{a}\|$, and

$$\|\text{proj}_a \mathbf{b}\| = -\|\mathbf{b}\| \cos \theta$$



Projection vectors

- When $0 \leq \theta < \frac{\pi}{2}$, $\text{proj}_a \mathbf{b}$ is in the direction of the unit vector $\mathbf{a}/\|\mathbf{a}\|$, and

$$\|\text{proj}_a \mathbf{b}\| = \|\mathbf{b}\| \cos \theta.$$

- When $\theta = \frac{\pi}{2}$,

$$\text{proj}_a \mathbf{b} = \mathbf{0}.$$

- When $\frac{\pi}{2} < \theta \leq \pi$, $\text{proj}_a \mathbf{b}$ is in the direction of the unit vector $-\mathbf{a}/\|\mathbf{a}\|$, and

$$\|\text{proj}_a \mathbf{b}\| = -\|\mathbf{b}\| \cos \theta$$

- We know that

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}.$$

- We have

$$\text{proj}_a \mathbf{b} = \|\mathbf{b}\| \cos \theta \frac{\mathbf{a}}{\|\mathbf{a}\|} = \|\mathbf{b}\| \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} \frac{\mathbf{a}}{\|\mathbf{a}\|} = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|^2} \mathbf{a}.$$

Projection vectors – example

Example

$$\mathbf{a} = (4, 0, -3), \quad \mathbf{b} = (3, 1, -7)$$

$$\begin{aligned} \text{proj}_{\mathbf{a}} \mathbf{b} &= \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|^2} \mathbf{a} = \frac{4 \times 3 + 0 \times 1 + (-3) \times (-7)}{4^2 + 0^2 + (-3)^2} \mathbf{a} \\ &= \frac{33}{25} (4, 0, -3) = \left(\frac{132}{25}, 0, -\frac{99}{25} \right). \end{aligned}$$

Vectors and matrices

- Definitions
- Vectors
- **Matrices**
- Proofs and principles

Matrices

- $\mathbb{R}$: the set of all real numbers

Definition

A *matrix with coefficients in $\mathbb{R}$* is a rectangular array where each *entry* is an element of $\mathbb{R}$.

Matrix A is said to have m *rows*, n *columns* and is of size $m \times n$.

$$A = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ a_{21} & \dots & a_{2n} \\ & \vdots & \\ a_{m1} & \dots & a_{mn} \end{pmatrix}.$$

We also write $A = (a_{ij})_{1 \leq i \leq m, 1 \leq j \leq n}$. When the size is clear from the context, we write $A = (a_{ij})$.

Matrices – examples

Example

- $A = \begin{pmatrix} 1 & 7.5 & 6 \\ 2 & 3 & 4 \end{pmatrix}$ has 2 rows, 3 columns and is of size 2×3 . $a_{11} = 1$, $a_{22} = 3$.
- $B = \begin{pmatrix} 4 & -2 \\ 1 & 7 \\ -5 & 3 \end{pmatrix}$ is of size 3×2 . $b_{12} = -2$, $b_{31} = -5$
- $C = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$ is of size 3×3
- $D = \begin{pmatrix} 7 \\ 1 \\ -2 \end{pmatrix}$ is a 3×1 matrix
- $E = (4, -3, 0)$ is a 1×3 matrix
- $F = (4)$ is a 1×1 matrix

Matrices

- The size of a matrix is always specified by stating the number of rows first. For example, a 3×4 matrix always has three rows and four columns, never four rows and three columns
- An $m \times n$ matrix can be thought of either as a collection of m row vectors, each having n coordinates, or as a collection of n column vectors, each having m coordinates.

Definition

Let $\mathcal{M}_{m \times n}$ denote the set of all $m \times n$ matrices

Rows and columns of matrices

$$A = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ a_{21} & \dots & a_{2n} \\ \vdots & & \\ a_{m1} & \dots & a_{mn} \end{pmatrix}$$

- a_{ij} denotes the entry in the i th row and j th column.
- The i th row of A is

$$(a_{i1}, \quad a_{i2}, \quad \dots \quad a_{in}).$$

- The j th column of A is

$$\begin{pmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{pmatrix}.$$

Rows and columns of matrices – Examples

Example

$$A = \begin{pmatrix} 1 & 7.5 & 6 \\ 2 & 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 3.5 & 7 \end{pmatrix}$$

- The 1st row of A is

$$(1, \ 7.5, \ 6).$$

- The 2nd column of B is

$$\begin{pmatrix} 0 \\ 7 \end{pmatrix}.$$

- $A \in \mathcal{M}_{2 \times 3}$, $B \in \mathcal{M}_{2 \times 2}$

Main diagonal of a matrix

$$A = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ a_{21} & \dots & a_{2n} \\ & \vdots & \\ a_{m1} & \dots & a_{mn} \end{pmatrix}$$

The *main diagonal* entries of A are $a_{11}, a_{22}, a_{33}, \dots$, those that lie on a diagonal line drawn down to the right, beginning from the upper-left corner of the matrix.

Example

$$A = \begin{pmatrix} 1 & 7.5 & 6 \\ 2 & 3 & 4 \end{pmatrix}$$

has main diagonal entries 1, 3.

Equal matrices

- Two matrices A and B are *equal* iff they have the same size and all corresponding entries are equal
- $A, B \in \mathcal{M}_{m \times n}$, then

$$A = B \iff a_{ij} = b_{ij}, \quad \forall i = 1, 2, \dots, m, \quad j = 1, 2, \dots, n$$

Zero matrices

A *zero matrix* is any matrix with all entries equal to 0, denoted O .

Example

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Square matrices

$$A = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ a_{21} & \dots & a_{2n} \\ \vdots & & \\ a_{n1} & \dots & a_{nn} \end{pmatrix}$$

An $n \times n$ matrix is called a *square matrix* (i.e. a matrix with the same number of rows and columns).

Example

$$A = \begin{pmatrix} 5 & 0 \\ 9 & -2 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

Diagonal matrices

$$A = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ a_{21} & \dots & a_{2n} \\ & \vdots & \\ a_{n1} & \dots & a_{nn} \end{pmatrix}$$

A square matrix of size $n \times n$ is a *diagonal matrix* if $a_{ij} = 0$ for $i \neq j$

Example

$$A = \begin{pmatrix} 6 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & -2 \end{pmatrix}, \quad B = \begin{pmatrix} 4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad C = \begin{pmatrix} -4 & 0 \\ 0 & 5 \end{pmatrix}$$

The following matrix is not a diagonal matrix

$$D = \begin{pmatrix} 4 & 3 \\ 0 & 1 \end{pmatrix}$$

Identity matrices

$$A = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ a_{21} & \dots & a_{2n} \\ & \vdots & \\ a_{n1} & \dots & a_{nn} \end{pmatrix}$$

An $n \times n$ *identity matrix*, normally denoted I_n , is a diagonal matrix whose diagonal entries are 1 and all other entries are 0, i.e. $a_{ii} = 1$ for $i = 1, 2, \dots, n$ and $a_{ij} = 0$ for $i \neq j$.

Example

$$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad I_4 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Upper triangular matrices

$$A = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ a_{21} & \dots & a_{2n} \\ & \vdots & \\ a_{n1} & \dots & a_{nn} \end{pmatrix}$$

An *upper triangular matrix* is a square matrix with all entries below the main diagonal equal to zero.

In other words, $A \in \mathcal{M}_{n \times n}$ is an *upper triangular matrix* if $a_{ij} = 0$ for $i > j$.

Example

$$P = \begin{pmatrix} 6 & 9 & 11 \\ 0 & -2 & 3 \\ 0 & 0 & 5 \end{pmatrix}, \quad U = \begin{pmatrix} 7 & -2 & 2 & 0 \\ 0 & -4 & 9 & 5 \\ 0 & 0 & 0 & 8 \\ 0 & 0 & 0 & 3 \end{pmatrix}$$

Lower triangular matrices

$$A = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ a_{21} & \dots & a_{2n} \\ & \vdots & \\ a_{n1} & \dots & a_{nn} \end{pmatrix}$$

An *lower triangular matrix* is a square matrix with all entries above the main diagonal equal to zero.

In other words, $A \in \mathcal{M}_{n \times n}$ is an *lower triangular matrix* if $a_{ij} = 0$ for $i < j$.

Example

$$L = \begin{pmatrix} 3 & 0 & 0 \\ 9 & -2 & 0 \\ 14 & -6 & 1 \end{pmatrix}$$

Transpose of a matrix

The *transpose* of $A \in \mathcal{M}_{m \times n}$, denoted $A^\top$, is the $n \times m$ matrix obtained by interchanging the rows and columns of A .

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ a_{21} & \cdots & a_{2n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}, \quad A^\top = \begin{pmatrix} a_{11} & \cdots & a_{m1} \\ a_{12} & \cdots & a_{m2} \\ \vdots & & \vdots \\ a_{1n} & \cdots & a_{mn} \end{pmatrix}.$$

Example

$$A = \begin{pmatrix} 1 & 7.5 & 6 \\ 2 & 3 & 4 \end{pmatrix}, \quad A^\top = \begin{pmatrix} 1 & 2 \\ 7.5 & 3 \\ 6 & 4 \end{pmatrix}.$$

A is of size 2×3 , $A^\top$ is of size 3×2 .

Note

$$(A^\top)^\top = A$$

Symmetric matrices

Definition

A square matrix $A \in \mathcal{M}_{n \times n}$ is *symmetric* if $A = A^\top$. It is *skew-symmetric* if $A = -A^\top$.

For a general matrix $A \in \mathcal{M}_{m \times n}$, we have $A^\top \in \mathcal{M}_{n \times m}$. Therefore, $A = A^\top$ or $A = -A^\top$ can hold only when $m = n$, so every symmetric or skew-symmetric matrix is square.

Example

$$A = \begin{pmatrix} 2 & 6 & 4 \\ 6 & -1 & 0 \\ 4 & 0 & -3 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & -1 & 3 & 6 \\ 1 & 0 & 2 & -5 \\ -3 & -2 & 0 & 4 \\ -6 & 5 & -4 & 0 \end{pmatrix}$$

A is symmetric and B is skew-symmetric

Matrix addition

Definition

Take $A = (a_{ij}), B = (b_{ij}) \in \mathcal{M}_{m \times n}$, the *sum* of A and B , $A + B$, is the $m \times n$ matrix whose (i, j) -entry is equal to $a_{ij} + b_{ij}$

Example

$$\begin{pmatrix} 6 & -3 & 2 \\ -7 & 0 & 4 \end{pmatrix} + \begin{pmatrix} 5 & -6 & -3 \\ -4 & -2 & -4 \end{pmatrix} = \begin{pmatrix} 11 & -9 & -1 \\ -11 & -2 & 0 \end{pmatrix}$$

Matrix addition – example

Example

Notice that the definition does not allow addition of matrices with different sizes.

$$A = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \quad B = (4, \ 5, \ 6).$$

We cannot add those two matrices together. But

$$A + B^{\top} = \begin{pmatrix} 1 + 4 \\ 2 + 5 \\ 3 + 6 \end{pmatrix} = \begin{pmatrix} 5 \\ 7 \\ 9 \end{pmatrix}$$

$$A^{\top} + B = (1, \ 2, \ 3) + (4, \ 5, \ 6) = (1 + 4, \ 2 + 5, \ 3 + 6) = (5, \ 7, \ 9).$$

Multiply a matrix by a scalar

Definition

Let $A = (a_{ij}) \in \mathcal{M}_{m \times n}$ and $\alpha \in \mathbb{R}$. The scalar multiple of A by α is $\alpha A \in \mathcal{M}_{m \times n}$ whose (i, j) -entry is equal to αa_{ij} .

Example

- $\alpha = -2$

$$A = \begin{pmatrix} 4 & -1 & 6 & 7 \\ 2 & 4 & 9 & -5 \end{pmatrix}, \quad -2A = \begin{pmatrix} -8 & 2 & -12 & -14 \\ -4 & -8 & -18 & 10 \end{pmatrix}$$

- $0A = O$ for any matrix A

Subtraction of matrices

- Let $-A$ denote the matrix $-1A$, the scalar multiple of A by -1
- We define subtraction of matrices as

$$A - B = A + (-B)$$

Example

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$$

$$A - B = A + (-B) = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} -5 & -6 \\ -7 & -8 \end{pmatrix} = \begin{pmatrix} 1-5 & 2-6 \\ 3-7 & 4-8 \end{pmatrix} = \begin{pmatrix} -4 & -4 \\ -4 & -4 \end{pmatrix}$$

Fundamental properties of addition and scalar multiplication

Theorem

For any matrices $A, B, C \in \mathcal{M}_{m \times n}$ and any scalars $\alpha, \beta \in \mathbb{R}$, we have

- | | |
|---|--|
| 1. $A + B = B + A$ | <i>Commutative law of addition</i> |
| 2. $A + (B + C) = (A + B) + C$ | <i>Associative law of addition</i> |
| 3. $O + A = A + O = A$ | <i>Existence of identity element for addition</i> |
| 4. $A + (-A) = (-A) + A = O$ | <i>Existence of inverse elements for addition</i> |
| 5. $\alpha(A + B) = \alpha A + \alpha B$ | <i>Distributive laws of scalar multiplication</i> |
| 6. $(\alpha + \beta)A = \alpha A + \beta A$ | <i>over matrix addition</i> |
| 7. $(\alpha\beta)A = \alpha(\beta A)$ | <i>Associativity of scalar multiplication</i> |
| 8. $1A = A$ | <i>Identity property for scalar multiplication</i> |

Transpose of matrices

Theorem

For any $A, B \in \mathcal{M}_{m \times n}$, $\alpha \in \mathbb{R}$, we have

1. $(A + B)^\top = A^\top + B^\top$
2. $(A - B)^\top = A^\top - B^\top$
3. $(\alpha A)^\top = \alpha A^\top$

Matrix multiplication

Definition

The product of $A \in \mathcal{M}_{m \times n}$ and $B \in \mathcal{M}_{n \times r}$ is the matrix $C = AB \in \mathcal{M}_{m \times r}$ whose (i, j) -entry is the dot product of the i th row of A with the j th column of B

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{i1} & a_{i2} & a_{i3} & \cdots & a_{in} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1j} & \cdots & b_{1r} \\ b_{21} & b_{22} & \cdots & b_{2j} & \cdots & b_{2r} \\ b_{31} & b_{32} & \cdots & b_{3j} & \cdots & b_{3r} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nj} & \cdots & b_{nr} \end{pmatrix} = \begin{pmatrix} c_{11} & c_{12} & \cdots & c_{1j} & \cdots & c_{1r} \\ c_{21} & c_{22} & \cdots & c_{2j} & \cdots & c_{2r} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ c_{i1} & c_{i2} & \cdots & c_{ij} & \cdots & c_{ir} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ c_{m1} & c_{m2} & \cdots & c_{mj} & \cdots & c_{mr} \end{pmatrix}$$

where

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj} = \sum_{k=1}^n a_{ik}b_{kj}.$$

Matrix multiplication

- Two matrices A, B can be multiplied (in that order) only if the number of columns of A is equal to the number of rows of B
- This ensures that each row of A contains the same number of entries as each column of B . Thus it is possible to perform the dot products needed to calculate C

Note

The dot product of two vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ is the same as the product of $\mathbf{a} \in \mathcal{M}_{1 \times n}$ and $\mathbf{b}^\top \in \mathcal{M}_{n \times 1}$

Example

$$(1, \ 2, \ 3) \begin{pmatrix} 4 \\ 5 \\ -6 \end{pmatrix} = 1 \times 4 + 2 \times 5 + 3 \times (-6) = 4 + 10 - 18 = -4$$

Matrix multiplication – example

Example

$$A = \begin{pmatrix} 5 & -1 & 4 \\ -3 & 6 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 9 & 4 & -8 & 2 \\ 7 & 6 & -1 & 0 \\ -2 & 5 & 3 & -4 \end{pmatrix}$$

$A \in \mathcal{M}_{2 \times 3}$, $B \in \mathcal{M}_{3 \times 4}$, A and B can be multiplied and the product $C \in \mathcal{M}_{2 \times 4}$.

To calculate c_{11} , we compute the dot product of the 1st row of A and the 1st column of B :

$$c_{11} = (5, -1, 4) \cdot (9, 7, -2) = 5 \times 9 + (-1) \times 7 + 4 \times (-2) = 45 - 7 - 8 = 30.$$

$$c_{23} = (-3, 6, 0) \cdot (-8, -1, 3) = (-3) \times (-8) + 6 \times (-1) + 0 \times 3 = 24 - 6 = 18$$

The other entries are computed similarly, we have

$$C = AB = \begin{pmatrix} 30 & 34 & -27 & -6 \\ 15 & 24 & 18 & -6 \end{pmatrix}$$

Identity matrix

Let $A \in \mathcal{M}_{m \times n}$ be any matrix, $I_n \in \mathcal{M}_{n \times n}$ and $I_m \in \mathcal{M}_{m \times m}$ be identity matrices. We have

$$AI_n = I_m A = A$$

Example

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}, \quad AI_3 = I_2 A = A$$

Transpose of matrix product

Theorem

Let $A \in \mathcal{M}_{m \times n}$, $B \in \mathcal{M}_{n \times r}$, then

$$(AB)^{\top} = B^{\top} A^{\top}$$

Example

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$(AB)^{\top} = B^{\top} A^{\top} = \begin{pmatrix} 2 & 5 \\ 1 & 4 \end{pmatrix}$$

Matrix multiplication – example

Example

$$D = \begin{pmatrix} -2 & 1 \\ 0 & 5 \\ 4 & -3 \end{pmatrix}, E = \begin{pmatrix} 1 & -6 \\ 0 & 2 \end{pmatrix}, F = (-4, 2, 1), G = \begin{pmatrix} 7 \\ -1 \\ 5 \end{pmatrix}, H = \begin{pmatrix} 5 & 0 \\ 1 & -3 \end{pmatrix}$$

The order in which multiplication is performed is important. Given two matrices

- Neither product may be defined (e.g. DG , GD).
- One product may be defined but not the other (e.g. ED is not defined)
- Both products may be defined, but the resulting sizes may differ (e.g. F , G)
- Both products may be defined, and the resulting sizes may agree, but the entries may differ (e.g. E and H)

$$DE = \begin{pmatrix} -2 & 14 \\ 0 & 10 \\ 4 & -30 \end{pmatrix}, GF = \begin{pmatrix} -28 & 14 & 7 \\ 4 & -2 & -1 \\ -20 & 10 & 5 \end{pmatrix}, FG = (-25), EH = \begin{pmatrix} -1 & 18 \\ 2 & -6 \end{pmatrix}, HE = \begin{pmatrix} 5 & -30 \\ 1 & -12 \end{pmatrix}$$

Fundamental properties of matrix multiplication

Theorem

For any matrices A, B, C where the following operations are well-defined, and for any scalars $\alpha \in \mathbb{R}$, we have

- | | |
|-------------------------------|--|
| 1. $A(BC) = (AB)C$ | <i>Associative law of multiplication</i> |
| 2. $A(B + C) = AB + AC$ | <i>Distributive law of matrix multiplication</i> |
| 3. $(A + B)C = AC + BC$ | <i>over addition</i> |
| 4. $\alpha(AB) = A(\alpha B)$ | <i>Associative law of scalar and matrix multiplication</i> |

Proofs for 2, 3, 4 are easy - compute both sides, show they are equal. Proof for 1 can be found in the book Appendix A.

Distributive law of matrix multiplication – example

Example

$$A = \begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} -2 & 0 \\ 4 & -1 \end{pmatrix}, \quad C = \begin{pmatrix} 3 & -2 \\ 1 & 0 \end{pmatrix}$$

$$B + C = \begin{pmatrix} -2 + 3 & 0 + (-2) \\ 4 + 1 & -1 + 0 \end{pmatrix} = \begin{pmatrix} 1 & -2 \\ 5 & -1 \end{pmatrix}$$

$$A(B+C) = \begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix} \cdot \begin{pmatrix} 1 & -2 \\ 5 & -1 \end{pmatrix} = \begin{pmatrix} 1 \cdot 1 + 2 \cdot 5 & 1 \cdot (-2) + 2 \cdot (-1) \\ -1 \cdot 1 + 3 \cdot 5 & -1 \cdot (-2) + 3 \cdot (-1) \end{pmatrix} = \begin{pmatrix} 11 & -4 \\ 14 & -1 \end{pmatrix}$$

$$AB = \begin{pmatrix} -2 + 8 & 0 + (-2) \\ 2 + 12 & 0 - 3 \end{pmatrix} = \begin{pmatrix} 6 & -2 \\ 14 & -3 \end{pmatrix}, \quad AC = \begin{pmatrix} 5 & -2 \\ 0 & 2 \end{pmatrix}$$

We have

$$AB + AC = \begin{pmatrix} 11 & -4 \\ 14 & -1 \end{pmatrix} = A(B + C)$$

Remark

Continue from the previous example

$$A = \begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} -2 & 0 \\ 4 & -1 \end{pmatrix}, \quad C = \begin{pmatrix} 3 & -2 \\ 1 & 0 \end{pmatrix}$$

$$B + C = \begin{pmatrix} -2 + 3 & 0 + (-2) \\ 4 + 1 & -1 + 0 \end{pmatrix} = \begin{pmatrix} 1 & -2 \\ 5 & -1 \end{pmatrix}$$

$$A(B + C) = AB + AC = \begin{pmatrix} 11 & -4 \\ 14 & -1 \end{pmatrix}$$

$$(B + C)A = \begin{pmatrix} 1 & -2 \\ 5 & -1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix} = \begin{pmatrix} 1 + 2 & 2 - 6 \\ 5 + 1 & 10 - 3 \end{pmatrix} = \begin{pmatrix} 3 & -4 \\ 6 & 7 \end{pmatrix}$$

We note that

$$A(B + C) \neq (B + C)A$$

Cancellation laws do not hold

We note that if $AB = AC$ and $A \neq O$, it does not necessarily follow that $B = C$. For example

$$A = \begin{pmatrix} 2 & 1 \\ 6 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} -1 & 0 \\ 5 & 2 \end{pmatrix}, \quad C = \begin{pmatrix} 3 & 1 \\ -3 & 0 \end{pmatrix}$$

$$AB = \begin{pmatrix} 3 & 2 \\ 9 & 6 \end{pmatrix}, \quad AC = \begin{pmatrix} 3 & 2 \\ 9 & 6 \end{pmatrix}$$

Similarly, if $BA = CA$, $A \neq O$, it does not necessarily follow that $B = C$

Linear combination of matrices

Definition

Given $A_1, A_2, \dots, A_r \in \mathcal{M}_{m \times n}$, $\alpha_1, \alpha_2, \dots, \alpha_r \in \mathbb{R}$, an expression of the form

$$\alpha_1 A_1 + \alpha_2 A_2 + \dots + \alpha_r A_r$$

is called a *linear combination* of $A_1, A_2, \dots, A_r$ with coefficients $\alpha_1, \alpha_2, \dots, \alpha_r$.

Linear combination of matrices

Let

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

Then

$$A\mathbf{x} = \begin{pmatrix} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \end{pmatrix} = x_1 \begin{pmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{pmatrix} + x_2 \begin{pmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{pmatrix} + \cdots + x_n \begin{pmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{pmatrix}$$

Linear combination of matrices

Observation

Given $A \in \mathcal{M}_{m \times n}$, $\mathbf{x} \in \mathcal{M}_{n \times 1}$ (equivalently, $\mathbf{x}^\top \in \mathbb{R}^n$ under our row-vector convention), the product $A\mathbf{x}$ can be expressed as a linear combination of the columns of A in which the coefficients are the entries of $\mathbf{x}$.

Example

The matrix product

$$\begin{pmatrix} -1 & 3 & 2 \\ 1 & 2 & -3 \\ 2 & 1 & -2 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ -9 \\ -3 \end{pmatrix}$$

can be written as

$$2 \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} - 1 \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} + 3 \begin{pmatrix} 2 \\ -3 \\ -2 \end{pmatrix} = \begin{pmatrix} 1 \\ -9 \\ -3 \end{pmatrix}$$

Linear combination of matrices and matrix product

Consider

$$AB = \begin{pmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{pmatrix} \begin{pmatrix} 4 & 1 & 4 & 3 \\ 0 & -1 & 3 & 1 \\ 2 & 7 & 5 & 2 \end{pmatrix} = \begin{pmatrix} 12 & 27 & 30 & 13 \\ 8 & -4 & 26 & 12 \end{pmatrix}$$

It follows from the above observation that the j th column of AB can be expressed as a linear combination of the columns of A in which the coefficients in the linear combination are the entries from the j th column of B

$$\begin{pmatrix} 12 \\ 8 \end{pmatrix} = 4 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + 0 \begin{pmatrix} 2 \\ 6 \end{pmatrix} + 2 \begin{pmatrix} 4 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 27 \\ -4 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ 6 \end{pmatrix} + 7 \begin{pmatrix} 4 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 30 \\ 26 \end{pmatrix} = 4 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + 3 \begin{pmatrix} 2 \\ 6 \end{pmatrix} + 5 \begin{pmatrix} 4 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 13 \\ 12 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} 2 \\ 6 \end{pmatrix} + 2 \begin{pmatrix} 4 \\ 0 \end{pmatrix}$$

Linear combination of matrices and matrix product

$$AB = \begin{pmatrix} 1 & 2 & 4 \\ 2 & 6 & 0 \end{pmatrix} \begin{pmatrix} 4 & 1 & 4 & 3 \\ 0 & -1 & 3 & 1 \\ 2 & 7 & 5 & 2 \end{pmatrix} = \begin{pmatrix} 12 & 27 & 30 & 13 \\ 8 & -4 & 26 & 12 \end{pmatrix}$$

Similarly, the i th row of AB can be considered as linear combinations of the rows of B with coefficients given by the entries from the i th row of A

$$\begin{aligned} (12, \ 27, \ 30, \ 13) &= 1(4, \ 1, \ 4, \ 3) + 2(0, \ -1, \ 3, \ 1) + 4(2, \ 7, \ 5, \ 2) \\ (8, \ -4, \ 26, \ 12) &= 2(4, \ 1, \ 4, \ 3) + 6(0, \ -1, \ 3, \ 1) + 0(2, \ 7, \ 5, \ 2) \end{aligned}$$

The zero matrix

- $AO = OA = O$
- If $AB = O$, it is not necessarily true that $A = O$ or $B = O$. For example

$$A = \begin{pmatrix} 2 & 1 \\ 6 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} -1 & 2 \\ 2 & -4 \end{pmatrix}, \quad AB = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

Powers of a square matrix

- Square matrices are the only matrices that can be multiplied by themselves
- $A \in \mathcal{M}_{m \times n}$, AA can be computed iff $m = n$

Definition

For $A \in \mathcal{M}_{n \times n}$, the (*nonnegative*) powers of A are given by

$$A^0 = I_n, \quad A^1 = A, \quad A^k = A^{k-1}A \text{ for } k \geq 2.$$

Example

$$A = \begin{pmatrix} 2 & 1 \\ -4 & 3 \end{pmatrix}$$

$$A^2 = AA = \begin{pmatrix} 0 & 5 \\ -20 & 5 \end{pmatrix}, \quad A^3 = A^2A = \begin{pmatrix} -20 & 15 \\ -60 & -5 \end{pmatrix}.$$

Note

$A^2A = AA^2$ follows from the associative law

Special cases when $AB = BA$

- Take $A \in \mathcal{M}_{m \times n}$, $B \in \mathcal{M}_{p \times r}$
- For AB to be defined, we need $p = n$
- For BA to be defined, we need $r = m$
- Then

$$AB \in \mathcal{M}_{m \times m}, \quad BA \in \mathcal{M}_{n \times n}$$

- For $AB = BA$, we need $m = n$, i.e.

$$A, B \in \mathcal{M}_{n \times n}$$

are square matrices

- In case $AB = BA$, we say A and B *commute* or A *commutes* with B .

Some special cases when $AB = BA$

Take $A, B \in \mathcal{M}_{n \times n}$

- $n = 1$, $AB = BA$
- If $B = A$, then $AB = BA = A^2$
- If $A = O$ or $B = O$, then $AB = BA = O$
- If $\exists \alpha \in \mathbb{R}$ s.t. $A = \alpha I_n$, then according to the associative law of scalar and matrix multiplication, and the property of the identity matrix

$$BA = B(\alpha I_n) = \alpha(BI_n) = \alpha(I_n B) = (\alpha I_n)B = AB$$

- Similarly, if $\exists \alpha \in \mathbb{R}$ s.t. $B = \alpha I_n$, we have $AB = BA$

Example

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \quad AB = BA = \begin{pmatrix} 2 & 4 \\ 6 & 8 \end{pmatrix}$$

Trace of a square matrix

Definition

Let $A = (a_{ij}) \in \mathcal{M}_{n \times n}$ be a square matrix. The *trace* of A , denoted $\operatorname{tr}(A)$ is given by the sum of the main diagonal entries of A , i.e.

$$\operatorname{tr}(A) = a_{11} + a_{22} + \cdots + a_{nn} = \sum_{i=1}^n a_{ii}.$$

Example

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 5 & -3 & 4 \\ 1 & 1 & -2 \end{pmatrix}, \quad \operatorname{tr}(A) = 1 + (-3) + (-2) = -4.$$

Vectors and matrices

- Definitions
- Vectors
- Matrices
- Proofs and principles

Fundamental properties of vector addition and scalar multiplication

Theorem

Take any $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^n$, any $\alpha, \beta \in \mathbb{R}$, we have

- | | |
|--|--|
| 1. $\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a}$ | <i>Commutative law of addition</i> |
| 2. $\mathbf{a} + (\mathbf{b} + \mathbf{c}) = (\mathbf{a} + \mathbf{b}) + \mathbf{c}$ | <i>Associative law of addition</i> |
| 3. $\mathbf{0} + \mathbf{a} = \mathbf{a} + \mathbf{0} = \mathbf{a}$ | <i>Existence of identity element for addition</i> |
| 4. $\mathbf{a} + (-\mathbf{a}) = (-\mathbf{a}) + \mathbf{a} = \mathbf{0}$ | <i>Existence of inverse elements for addition</i> |
| 5. $\alpha(\mathbf{a} + \mathbf{b}) = \alpha\mathbf{a} + \alpha\mathbf{b}$ | <i>Distributive laws of scalar multiplication</i> |
| 6. $(\alpha + \beta)\mathbf{a} = \alpha\mathbf{a} + \beta\mathbf{a}$ | <i>over vector addition</i> |
| 7. $(\alpha\beta)\mathbf{a} = \alpha(\beta\mathbf{a})$ | <i>Associativity of scalar multiplication</i> |
| 8. $1\mathbf{a} = \mathbf{a}$ | <i>Identity property for scalar multiplication</i> |

- $\mathbf{0}$ is called an *identity element for vector addition* because $\mathbf{0}$ does not change the identity of any vector to which it is added
- $-\mathbf{a}$ is called the *Additive inverse of $\mathbf{a}$* because it “cancels out $\mathbf{a}$ ” to produce the additive identity element (i.e. the zero vector)

Proof of property 6

$$\begin{aligned}(\alpha + \beta)\mathbf{a} &= (\alpha + \beta)(a_1, a_2, \dots, a_n) \\&= ((\alpha + \beta)a_1, (\alpha + \beta)a_2, \dots, (\alpha + \beta)a_n) \\&\quad \text{definition of scalar multiplication} \\&= (\alpha a_1 + \beta a_1, \alpha a_2 + \beta a_2, \dots, \alpha a_n + \beta a_n) \\&\quad \text{coordinate-wise use of distributive law in } \mathbb{R} \\&= (\alpha a_1, \alpha a_2, \dots, \alpha a_n) + (\beta a_1, \beta a_2, \dots, \beta a_n) \\&\quad \text{definition of vector addition} \\&= \alpha(a_1, a_2, \dots, a_n) + \beta(a_1, a_2, \dots, a_n) \\&\quad \text{definition of scalar multiplication} \\&= \alpha\mathbf{a} + \beta\mathbf{a}\end{aligned}$$

Properties of dot product

Theorem

Take any $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^n$, $\alpha \in \mathbb{R}$, then

1. $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$ *Commutativity of dot product*
2. $\mathbf{a} \cdot \mathbf{a} = \|\mathbf{a}\|^2 \geq 0$ *Relationship between dot product and norm*
3. $\mathbf{a} \cdot \mathbf{a} = 0$ iff $\mathbf{a} = \mathbf{0}$
4. $\alpha(\mathbf{a} \cdot \mathbf{b}) = (\alpha\mathbf{a}) \cdot \mathbf{b} = \mathbf{a} \cdot (\alpha\mathbf{b})$ *Relationship between scalar multiplication and dot product*
5. $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$ *Distributive laws of dot product*
6. $(\mathbf{a} + \mathbf{b}) \cdot \mathbf{c} = (\mathbf{a} \cdot \mathbf{c}) + (\mathbf{b} \cdot \mathbf{c})$ *over addition*

Proof.

We provide the proof for a few properties.

$$2. \mathbf{a} = (a_1, a_2, \dots, a_n)$$

$$\mathbf{a} \cdot \mathbf{a} = a_1a_1 + a_2a_2 + \dots + a_na_n = a_1^2 + a_2^2 + \dots + a_n^2 = \|\mathbf{a}\|^2 \geq 0$$

Properties of dot product

5. $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$ Distributive laws of dot product over addition

Proof.

$$5. \mathbf{a} = (a_1, a_2, \dots, a_n), \mathbf{b} = (b_1, b_2, \dots, b_n), \mathbf{c} = (c_1, c_2, \dots, c_n)$$

$$\begin{aligned} \mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) &= (a_1, a_2, \dots, a_n) \cdot ((b_1, b_2, \dots, b_n) + (c_1, c_2, \dots, c_n)) \\ &= (a_1, a_2, \dots, a_n) \cdot (b_1 + c_1, b_2 + c_2, \dots, b_n + c_n) \\ &= a_1(b_1 + c_1) + a_2(b_2 + c_2) + \dots + a_n(b_n + c_n) \\ &= a_1b_1 + a_1c_1 + a_2b_2 + a_2c_2 + \dots + a_nb_n + a_nc_n \\ &= (a_1b_1 + a_2b_2 + \dots + a_nb_n) + (a_1c_1 + a_2c_2 + \dots + a_nc_n) \end{aligned}$$

$$\begin{aligned} \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c} &= ((a_1, a_2, \dots, a_n) \cdot (b_1, b_2, \dots, b_n)) \\ &\quad + ((a_1, a_2, \dots, a_n) \cdot (c_1, c_2, \dots, c_n)) \\ &= (a_1b_1 + a_2b_2 + \dots + a_nb_n) + (a_1c_1 + a_2c_2 + \dots + a_nc_n) \end{aligned}$$

Dot product of unit vectors

Recall

A vector of norm 1 is called a *unit vector*

Lemma

If $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ are unit vectors, then

$$-1 \leq \mathbf{a} \cdot \mathbf{b} \leq 1.$$

Proof.

We make use of different properties of dot product

Dot product of unit vectors

Lemma

If $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ are unit vectors, then

$$-1 \leq \mathbf{a} \cdot \mathbf{b} \leq 1.$$

Proof.

$$\begin{aligned}(\mathbf{a} + \mathbf{b}) \cdot (\mathbf{a} + \mathbf{b}) &= \|\mathbf{a} + \mathbf{b}\|^2 \geq 0 && \text{property 2} \\ \implies \mathbf{a} \cdot \mathbf{a} + \mathbf{b} \cdot \mathbf{a} + \mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{b} &\geq 0 && \text{properties 5, 6} \\ \implies \|\mathbf{a}\|^2 + 2(\mathbf{a} \cdot \mathbf{b}) + \|\mathbf{b}\|^2 &\geq 0 && \text{properties 1, 2} \\ \implies 1 + 2\mathbf{a} \cdot \mathbf{b} + 1 &\geq 0 && \mathbf{a}, \mathbf{b} \text{ are unit vectors} \\ \implies \mathbf{a} \cdot \mathbf{b} &\geq -1\end{aligned}$$

A similar argument beginning with $(\mathbf{a} - \mathbf{b}) \cdot (\mathbf{a} - \mathbf{b}) = \|\mathbf{a} - \mathbf{b}\|^2 \geq 0$ shows $\mathbf{a} \cdot \mathbf{b} \leq 1$ $\square$

Fundamental properties of addition and scalar multiplication

Theorem

For any matrices $A, B, C \in \mathcal{M}_{m \times n}$ and any scalars $\alpha, \beta \in \mathbb{R}$, we have

- | | |
|---|--|
| 1. $A + B = B + A$ | <i>Commutative law of addition</i> |
| 2. $A + (B + C) = (A + B) + C$ | <i>Associative law of addition</i> |
| 3. $O + A = A + O = A$ | <i>Existence of identity element for addition</i> |
| 4. $A + (-A) = (-A) + A = O$ | <i>Existence of inverse elements for addition</i> |
| 5. $\alpha(A + B) = \alpha A + \alpha B$ | <i>Distributive laws of scalar multiplication</i> |
| 6. $(\alpha + \beta)A = \alpha A + \beta A$ | <i>over matrix addition</i> |
| 7. $(\alpha\beta)A = \alpha(\beta A)$ | <i>Associativity of scalar multiplication</i> |
| 8. $1A = A$ | <i>Identity property for scalar multiplication</i> |

To prove each property, calculate corresponding entries on both sides and show they agree by applying an appropriate law of real numbers.

Fundamental properties of addition and scalar multiplication

Theorem

$$1. A + B = B + A$$

Commutative law of addition

To prove each property, calculate corresponding entries on both sides and show they agree by applying an appropriate law of real numbers.

Proof.

Part 1. Suppose $A = (a_{ij})$, $B = (b_{ij})$, then the (i, j) -entry of $A + B$ is $a_{ij} + b_{ij}$. And the (i, j) -entry of $B + A$ is $b_{ij} + a_{ij}$. By the commutativity property of addition for real numbers, we have

$$a_{ij} + b_{ij} = b_{ij} + a_{ij},$$

which implies $A + B = B + A$. □

Transpose of matrices

Theorem

For any $A, B \in \mathcal{M}_{m \times n}$, $\alpha \in \mathbb{R}$, we have

1. $(A + B)^\top = A^\top + B^\top$
2. $(A - B)^\top = A^\top - B^\top$
3. $(\alpha A)^\top = \alpha A^\top$

Proof.

Proof of part 1. Suppose $A = (a_{ij})$, $B = (b_{ij})$. First we note that $(A + B)^\top, A^\top + B^\top \in \mathcal{M}_{n \times m}$. Next, we show that their (i, j) -entries are equal for $i = 1, 2, \dots, n$, $j = 1, 2, \dots, m$:

$$(i, j) - \text{entry of } (A + B)^\top = (j, i) - \text{entry of } A + B = a_{ji} + b_{ji}$$

$$(i, j) - \text{entry of } A^\top + B^\top = (j, i) - \text{entry of } A + (j, i) - \text{entry of } B = a_{ji} + b_{ji}$$



Identity matrix

Let $A \in \mathcal{M}_{m \times n}$ be any matrix, $I_n \in \mathcal{M}_{n \times n}$ and $I_m \in \mathcal{M}_{m \times m}$ be identity matrices. We have

$$AI_n = I_m A = A$$

Proof.

Suppose $I_n = (c_{ij})$, where $i, j = 1, 2, \dots, n$, and

$$c_{ij} = \begin{cases} 0 & i \neq j, \\ 1 & i = j. \end{cases}$$

Let $B = AI_n = (b_{ij})$. Then, for $i = 1, 2, \dots, m$ and $j = 1, 2, \dots, n$,

$$b_{ij} = \sum_{k=1}^n a_{ik} c_{kj} = a_{ij}.$$

Hence $AI_n = A$. Similarly, applying the same entrywise argument to $I_m A$ gives $I_m A = A$.

Transpose of matrix product

Theorem

Let $A \in \mathcal{M}_{m \times n}$, $B \in \mathcal{M}_{n \times r}$, then

$$(AB)^{\top} = B^{\top} A^{\top}$$

Proof.

First we note that both matrices are of size $r \times m$.

$$\begin{aligned} (i, j) - \text{entry of } (AB)^{\top} &= (j, i) - \text{entry of } AB \\ &= [j\text{th row of } A] \cdot [i\text{th column of } B] \\ (i, j) - \text{entry of } B^{\top} A^{\top} &= [i\text{th row of } B^{\top}] \cdot [j\text{th column of } A^{\top}] \\ &= [i\text{th column of } B] \cdot [j\text{th row of } A] \\ &= [j\text{th row of } A] \cdot [i\text{th column of } B] \end{aligned}$$



Properties of triangular matrices

Theorem

1. *The transpose of a lower triangular matrix is upper triangular, and the transpose of an upper triangular matrix is lower triangular.*
2. *The product of lower triangular matrices is lower triangular, and the product of upper triangular matrices is upper triangular.*

Proof.

- Part 1 is immediate. We prove part 2.
- Let $A = (a_{ij})$, $B = (b_{ij}) \in \mathcal{M}_{n \times n}$ be lower triangular. Suppose $C = (c_{ij}) = AB$.
- When $i < j$

$$c_{ij} = (a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{i(j-1)}b_{(j-1)j}) + (a_{ij}b_{jj} + \cdots + a_{in}b_{nj})$$

- In the first grouping of terms, b factors are zero since B is lower triangular
- In the second grouping of terms, a factors are zero since A is lower triangular